

# Model Question Paper- II with effect from 2022

## CBCS SCHEME

### Fourth Semester B.E Degree Examination 2024-25

#### Database Management System (BCS403)

TIME: 03 Hours

Max.Marks:100

1. Note: Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**

2. M: Marks, L: Bloom's level, C: Course outcomes.

|            | <b>Module - 1</b> |                                                                                                                               | <b>M</b>  | <b>L</b>  | <b>C</b>   |
|------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.1</b> | <b>a</b>          | What is a Database? Explain the three schema architecture with neat diagram.                                                  | <b>8</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>b</b>          | What are the advantages of using DBMS approach? Explain                                                                       | <b>8</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>c</b>          | Explain the following terms.<br>1. Data Dictionary 2. Weak Entity                                                             | <b>4</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>OR</b>         |                                                                                                                               |           |           |            |
| <b>Q.2</b> | <b>a</b>          | Explain the categories of Data Models.                                                                                        | <b>8</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>b</b>          | Explain the component modules of DBMS & their interactions with diagram.                                                      | <b>8</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>c</b>          | What are the responsibilities of DBA & database designers?                                                                    | <b>4</b>  | <b>L2</b> | <b>CO1</b> |
|            | <b>Module - 2</b> |                                                                                                                               |           |           |            |
| <b>Q.3</b> | <b>a</b>          | Explain the different types of update operations on relational database. How basic operation deals with constraint violation. | <b>6</b>  | <b>L2</b> | <b>CO2</b> |
|            | <b>b</b>          | Explain Unary relational operations with examples.                                                                            | <b>6</b>  | <b>L2</b> | <b>CO2</b> |
|            | <b>c</b>          | What is an Integrity Constraint? Explain the importance of Referential Integrity Constraint.                                  | <b>8</b>  | <b>L2</b> | <b>CO2</b> |
|            | <b>OR</b>         |                                                                                                                               |           |           |            |
| <b>Q.4</b> | <b>a</b>          | Explain the following relational algebra operation.<br>JOIN, DIFFERENCE, SELECT, UNION                                        | <b>10</b> | <b>L3</b> | <b>CO2</b> |
|            | <b>b</b>          | Discuss the E.R to Relational mapping algorithm with example for each step.                                                   | <b>6</b>  | <b>L3</b> | <b>CO2</b> |
|            | <b>c</b>          | Explain the relational algebra operation for set theory with examples.                                                        | <b>4</b>  | <b>L2</b> | <b>CO2</b> |
|            | <b>Module - 3</b> |                                                                                                                               |           |           |            |
| <b>Q.5</b> | <b>a</b>          | Illustrate insert, delete, update, alter & drop commands in SQL.                                                              | <b>6</b>  | <b>L4</b> | <b>CO3</b> |

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|             |                   |                                                                                                                                                                                             |           |           |            |
|-------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
|             | <b>b</b>          | Explain informal design guidelines for relational schema design.                                                                                                                            | <b>4</b>  | <b>L2</b> | <b>CO3</b> |
|             | <b>c</b>          | What is Functional dependency? Explain the inference rules for functional dependency with proof.                                                                                            | <b>10</b> | <b>L3</b> | <b>CO4</b> |
|             | <b>OR</b>         |                                                                                                                                                                                             |           |           |            |
| <b>Q.6</b>  | <b>a</b>          | Consider two sets of functional dependency. $F=\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ $E= \{A \rightarrow CD, E \rightarrow AH\}$ . Are they Equivalent? | <b>10</b> | <b>L3</b> | <b>CO4</b> |
|             | <b>b</b>          | Explain the types of update anomalies in SQL with an example.                                                                                                                               | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|             | <b>Module - 4</b> |                                                                                                                                                                                             |           |           |            |
| <b>Q.7</b>  | <b>a</b>          | Demonstrate transaction states & additional operations.                                                                                                                                     | <b>10</b> | <b>L3</b> | <b>CO4</b> |
|             | <b>b</b>          | Demonstrate working of Assertion & Triggers in database? Explain with an example.                                                                                                           | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|             | <b>OR</b>         |                                                                                                                                                                                             |           |           |            |
| <b>Q.8</b>  | <b>a</b>          | Demonstrate the System Log in database transaction.                                                                                                                                         | <b>6</b>  | <b>L2</b> | <b>CO4</b> |
|             | <b>b</b>          | Discuss the ACID properties of database transaction.                                                                                                                                        | <b>4</b>  | <b>L2</b> | <b>CO4</b> |
|             | <b>c</b>          | Explain stored procedure language in SQL with an example.                                                                                                                                   | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|             | <b>Module - 5</b> |                                                                                                                                                                                             |           |           |            |
| <b>Q.9</b>  | <b>a</b>          | Explain the Two phase locking protocol used for concurrency control.                                                                                                                        | <b>8</b>  | <b>L3</b> | <b>CO5</b> |
|             | <b>b</b>          | Define Schedule? Illustrate with an example.                                                                                                                                                | <b>4</b>  | <b>L2</b> | <b>CO5</b> |
|             | <b>c.</b>         | Why Concurrency control is needed? Demonstrate with an example.                                                                                                                             | <b>8</b>  | <b>L3</b> | <b>CO5</b> |
|             | <b>OR</b>         |                                                                                                                                                                                             |           |           |            |
| <b>Q.10</b> | <b>a</b>          | What is NOSQL? Explain the CAP theorem.                                                                                                                                                     | <b>6</b>  | <b>L2</b> | <b>CO5</b> |
|             | <b>b</b>          | What are document based NOSQL systems? basic operations CRUD in MongoDB.                                                                                                                    | <b>8</b>  | <b>L2</b> | <b>CO5</b> |
|             | <b>c</b>          | What is NOSQL Graph database? Explain Neo4j.                                                                                                                                                | <b>6</b>  | <b>L2</b> | <b>CO5</b> |

# Model Question Paper- I with effect from 2022

## CBCS SCHEME

### Fourth Semester B.E Degree Examination 2024-25

#### Database Management Systems (BCS403)

TIME: 03 Hours

Max.Marks:100

1. Note: Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**  
2. M: Marks, L: Bloom's level, C: Course outcomes.

|     | Module - 1 |                                                                                                                        | M | L  | C   |
|-----|------------|------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1 | a          | Explain the types of end users with examples.                                                                          | 8 | L2 | CO1 |
|     | b          | What are the advantages of using DBMS? Explain.                                                                        | 8 | L2 | CO1 |
|     | c          | Describe the characteristics of database.                                                                              | 4 | L2 | CO1 |
|     | OR         |                                                                                                                        |   |    |     |
| Q.2 | a          | Explain three schema architecture. Why mappings b/w schema levels are required?                                        | 8 | L2 | CO1 |
|     | b          | Explain the different types of attributes in ER model.                                                                 | 8 | L2 | CO1 |
|     | c          | Explain the following.<br>1. Cardinality Ratio    2. Weak Entity                                                       | 4 | L2 | CO1 |
|     | Module - 2 |                                                                                                                        |   |    |     |
| Q.3 | a          | Explain the different Relational Model constraints.                                                                    | 6 | L2 | CO2 |
|     | b          | Demonstrate the concepts of Generalization & Specialization with examples.                                             | 6 | L2 | CO2 |
|     | c          | Explain Entity Integrity Constraint & Referential Integrity Constraints? Why each of these is important in a database. | 8 | L2 | CO2 |
|     | OR         |                                                                                                                        |   |    |     |

## Model Question Paper- I with effect from 2022

|                   |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |            |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.4</b>        | <b>a</b> | Consider the Sailors-Boats-Reserves DB described<br><br>s (sid, sname, rating, age)<br><br>b (bid, bname, color)<br><br>r (sid, bid, date)<br><br>Write each of the following queries in SQL.<br><br>1. Find the colors of boats reserved by Alber.<br><br>2. Find all sailor ids of sailors who have a rating of at least 8 or reserved boat 103.<br><br>3. Find the names of sailors who have not reserved a boat whose name contains the string "storm". Order the names in ascending order.<br><br>4. Find the sailor ids of sailors with age over 20 who have not reserved a boat whose name includes the string "thunder". | <b>10</b> | <b>L3</b> | <b>CO2</b> |
|                   | <b>b</b> | Discuss the Equijoin & Natural Join with suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>6</b>  | <b>L3</b> | <b>CO2</b> |
|                   | <b>c</b> | Explain the relational algebra operation for set theory with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>4</b>  | <b>L2</b> | <b>CO2</b> |
| <b>Module - 3</b> |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.5</b>        | <b>a</b> | Explain the Cursor & its properties in embedded SQL with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>6</b>  | <b>L2</b> | <b>CO3</b> |
|                   | <b>b</b> | What is a Normalization? Explain the 1NF, 2NF & 3NF with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>c</b> | Explain informal design guidelines for relational schema design.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>4</b>  | <b>L2</b> | <b>CO3</b> |
| <b>OR</b>         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.6</b>        | <b>a</b> | What is Functional Dependency? Write algorithm to find minimal cover for set of Functional Dependency. Construct the minimal cover m for set of functional dependency. E={ B→A, D→A, AB→D }                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b</b> | Explain the types of update anomalies in SQL with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>10</b> | <b>L4</b> | <b>CO3</b> |
| <b>Module - 4</b> |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.7</b>        | <b>a</b> | Demonstrate the Database Transaction with transaction diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>10</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b</b> | Demonstrate working of Assertion & Triggers in SQL? Explain with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>10</b> | <b>L3</b> | <b>CO3</b> |
| <b>OR</b>         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.8</b>        | <b>a</b> | Demonstrate the System Log in database transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>6</b>  | <b>L2</b> | <b>CO4</b> |
|                   | <b>b</b> | Demonstrate the ACID properties of database transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>4</b>  | <b>L2</b> | <b>CO4</b> |
|                   | <b>c</b> | Explain stored procedure language in SQL with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>10</b> | <b>L2</b> | <b>CO3</b> |

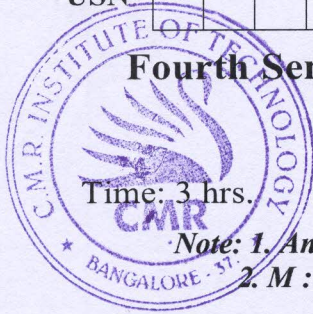
## Model Question Paper- I with effect from 2022

|             |                   |                                                                                  |          |           |            |
|-------------|-------------------|----------------------------------------------------------------------------------|----------|-----------|------------|
|             | <b>Module - 5</b> |                                                                                  |          |           |            |
| <b>Q.9</b>  | <b>a</b>          | Demonstrate the Two phase locking protocol used for concurrency control.         | <b>8</b> | <b>L3</b> | <b>CO5</b> |
|             | <b>b</b>          | Demonstrate the Concurrency control based on Timestamp ordering.                 | <b>4</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>c.</b>         | Why Concurrency control is needed? Demonstrate with an example.                  | <b>8</b> | <b>L3</b> | <b>CO5</b> |
|             | <b>OR</b>         |                                                                                  |          |           |            |
| <b>Q.10</b> | <b>a</b>          | What is NOSQL? Explain the CAP theorem.                                          | <b>6</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>b</b>          | What are document based NOSQL systems? Explain basic operations CRUD in MongoDB. | <b>8</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>c</b>          | What is NOSQL Graph database? Explain Neo4j.                                     | <b>6</b> | <b>L2</b> | <b>CO5</b> |

# CBCS SCHEME

USN

BCS403



## Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Database Management Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks, L: Bloom's level, C: Course outcomes.*

| Module – 1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       | M     | L     | C   |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-----|--------|------|--------|--------|---------|-------|-----|------|-------|------|--------|---------|-------|-------|------|-------|---------|--------|-------|-----|---------|---------|--------|------|----|----|-----|
| Q.1        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Define database. Elaborate component modules of DBMS and their interactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       | 10    | L2    | CO1 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Describe the three-schema architecture. Why do we need mappings among schema levels?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       | 06    | L2    | CO1 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|            | c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Explain the difference between logical and physical data independence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       | 04    | L2    | CO1 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| OR         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Q.2        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Draw an ER diagram for an COMPANY database with employee, department, project as strong entities and dependent as weak entity. Specify the constraints, relationships and ratios in the ER diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       | 10    | L3    | CO3 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Define the following terms with example for each using ER notations: Entity, attribute, composite attribute, multivalued attribute, participation role.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       | 10    | L3    | CO3 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Module – 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Q.3        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Discuss the update operations and dealing with constraint violations with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       | 08    | L2    | CO2 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Illustrate the relational algebra operators with examples for select and project operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       | 06    | L2    | CO2 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
|            | c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Discuss the characteristics of relations that make them different from ordinary table and files.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       | 06    | L2    | CO2 |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| OR         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Q.4        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Perform (i) Student U instructor (ii) Student ∩ Instructor (iii) Student – Instructor (iv) Instructor – Student on the following tables:<br><div><div>Student</div><table><tr><td>Fname</td><td>Lname</td></tr><tr><td>Susan</td><td>Yao</td></tr><tr><td>Ramesh</td><td>Shah</td></tr><tr><td>Johnny</td><td>Kohler</td></tr><tr><td>Barbara</td><td>Jones</td></tr><tr><td>Amy</td><td>Ford</td></tr><tr><td>Jimmy</td><td>Wang</td></tr><tr><td>Ernest</td><td>Gilbert</td></tr></table><div>Instructor</div><table><tr><td>Fname</td><td>Lname</td></tr><tr><td>John</td><td>Smith</td></tr><tr><td>Ricardo</td><td>Browne</td></tr><tr><td>Susan</td><td>Mao</td></tr><tr><td>Francis</td><td>Johnson</td></tr><tr><td>Ramesh</td><td>Shah</td></tr></table></div> | Fname | Lname | Susan | Yao | Ramesh | Shah | Johnny | Kohler | Barbara | Jones | Amy | Ford | Jimmy | Wang | Ernest | Gilbert | Fname | Lname | John | Smith | Ricardo | Browne | Susan | Mao | Francis | Johnson | Ramesh | Shah | 04 | L3 | CO2 |
|            | Fname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Lname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Susan      | Yao                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Ramesh     | Shah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Johnny     | Kohler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Barbara    | Jones                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Amy        | Ford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Jimmy      | Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Ernest     | Gilbert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Fname      | Lname                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| John       | Smith                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Ricardo    | Browne                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Susan      | Mao                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Francis    | Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| Ramesh     | Shah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |       |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |
| b.         | Consider the following relational database schema and write the queries in relational algebra expressions:<br>EMP(Eno, Ename, Salary, Address, Phone, DNo)<br>DEPT(DNo, Dname, DLoc, MgrEno)<br>DEPENDENT(Eno, Dep_Name, Drelation, Dage)<br>(i) List all the employees who reside in ‘Belagavi’.<br>(ii) List all the employees who earn salary between 30000 and 40000<br>(iii) List all the employees who work for the ‘Sales’ department<br>(iv) List all the employees who have at least one daughter<br>(v) List the department names along with the names of the managers |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10    | L3    | CO2   |     |        |      |        |        |         |       |     |      |       |      |        |         |       |       |      |       |         |        |       |     |         |         |        |      |    |    |     |

| c.         | Consider the two tables $T_1$ and $T_2$ shown below:<br><div><div><math>T_1</math><table><tr><th>P</th><th>Q</th><th>R</th></tr><tr><td>10</td><td>a</td><td>5</td></tr><tr><td>15</td><td>b</td><td>8</td></tr><tr><td>25</td><td>a</td><td>6</td></tr></table></div><div><math>T_2</math><table><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>10</td><td>b</td><td>6</td></tr><tr><td>25</td><td>c</td><td>3</td></tr><tr><td>10</td><td>b</td><td>5</td></tr></table></div></div><br>Show the results of the following operations:<br>(i) $T_1 \bowtie_{T_1.P=T_2.A} T_2$<br>(ii) $T_1 \bowtie_{T_1.Q=T_2.B} T_2$<br>(iii) $T_1 \bowtie_{(T_1.P=T_2.A \text{ AND } T_1.R=T_2.C)} T_2$ | P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Q  | R  | 10  | a | 5 | 15 | b | 8 | 25 | a | 6 | A | B | C | 10 | b | 6 | 25 | c | 3 | 10 | b | 5 | 06 | L3 | CO2 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|---|---|----|---|---|----|---|---|---|---|---|----|---|---|----|---|---|----|---|---|----|----|-----|
| P          | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 10         | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 15         | b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 25         | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| A          | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 10         | b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 25         | c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| 10         | b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Module – 3 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Q.5        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Discuss the informal design guidelines for relation schema design.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 08 | L2 | CO4 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Define 1NF, 2NF, and 3NF with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 06 | L2 | CO4 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Write the syntax for INSERT, UPDATE and DELETE statements in SQL and explain with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 06 | L2 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| OR         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Q.6        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Discuss insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10 | L2 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Illustrate the following with suitable examples:<br>(i) Datatypes in SQL<br>(ii) Substring Pattern Matching in SQL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10 | L2 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Module – 4 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Q.7        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Consider the following relations:<br>Student( <u>Snum</u> , Sname, Branch, level, age)<br>Class( <u>Cname</u> , meet_at, room, fid)<br>Enrolled( <u>Snum</u> , <u>Cname</u> )<br>Faculty( <u>fid</u> , fname, deptid)<br>Write the following queries in SQL. No duplicates should be printed in any of the answers.<br>(i) Find the names of all Juniors (level = JR) who are enrolled in a class taught by I. Teach.<br>(ii) Find the names of all classes that either meet in room R128 or have five or more students enrolled.<br>(iii) For all levels except JR, print the level and the average age of students for that level.<br>(iv) For each faculty member that has taught classes only in room R128, print the faculty member's name and the total number of classes she or he has taught.<br>(v) Find the names of students not enrolled in any class. | 10 | L3 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | What do you understand by correlated Nested Queries in SQL? Explain with suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 04 | L2 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Discuss the ACID properties of a database transaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 06 | L2 | CO4 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| OR         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
| Q.8        | a.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | What are the views in SQL? Explain with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 04 | L3 | CO5 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | b.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | In SQL, write the usage of GROUP BY and HAVING clauses with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 06 | L2 | CO3 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |
|            | c.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Discuss the types of problems that may encounter with transactions that run concurrently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 | L2 | CO5 |   |   |    |   |   |    |   |   |   |   |   |    |   |   |    |   |   |    |   |   |    |    |     |

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## Module – 5

|      |    |                                                                                                                  |    |    |     |
|------|----|------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.9  | a. | What is the two phase locking protocol? How does it Guarantee serializability.                                   | 06 | L2 | CO5 |
|      | b. | Describe the wait-die and wound-wait protocols for deadlock prevention.                                          | 08 | L2 | CO5 |
|      | c. | List and explain the four major categories of NOSQL system.                                                      | 06 | L2 | CO3 |
| OR   |    |                                                                                                                  |    |    |     |
| Q.10 | a. | What is Multiple Granularity locking? How is it implemented using intension locks? Explain.                      | 10 | L2 | CO5 |
|      | b. | Discuss the following MongoDB CRUD operations with their formats:<br>(i) Insert      (ii) Delete      (iii) Read | 06 | L2 | CO4 |
|      | c. | Briefly discuss about Neo4j data model.                                                                          | 04 | L2 | CO4 |

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**Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025**  
**Database Management Systems**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
 2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Explain the types of attributes with example.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 | L2 | CO1 |
|            | b. | Define database. Explain the main characteristics of the database approach.                                                                                                                                                                                                                                                                                                                                                                                                                              | 8 | L2 | CO1 |
|            | c. | Show the ER diagram for an EMPLOYEE database by assuming your own entities (minimum 4) attributes and relationships, mention cardinality ratios wherever appropriate.                                                                                                                                                                                                                                                                                                                                    | 8 | L3 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |     |
| Q.2        | a. | Describe the three schema architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 | L2 | CO1 |
|            | b. | Explain the component models of DBMS and their interaction with the help of diagram.                                                                                                                                                                                                                                                                                                                                                                                                                     | 8 | L2 | CO1 |
|            | c. | Design ER diagram for a university database by assuming your own entities (4). Mention primary key , constraints and relationships.                                                                                                                                                                                                                                                                                                                                                                      | 8 | L3 | CO2 |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |     |
| Q.3        | a. | Explain relational model constraints.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6 | L2 | CO1 |
|            | b. | Explain the characteristics of relations with suitable example for each.                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6 | L2 | CO1 |
|            | c. | Considering the following schema :<br>Sailors ( <u>sid</u> , sname , rating , age)<br>Boats ( <u>bid</u> , bname , color)<br>Reserves ( <u>sid</u> , bid , day)<br>Write a relational algebra queries for the following :<br>i) Find the names of sailors, who have reserved red and a green boat.<br>ii) Find the names of sailors who have reserved a red boat.<br>iii) Find the names of sailors who have reserved a red or green boat.<br>iv) Find the names of sailors who have reserved all boats. | 8 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |     |
| Q.4        | a. | Explain the steps to convert the basic ER model to relational Database schema.                                                                                                                                                                                                                                                                                                                                                                                                                           | 6 | L2 | CO1 |
|            | b. | Explain Unary relational operations with example.                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6 | L2 | CO1 |

|                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
|----------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|-----|---|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---|----|-----|
|                | c.             | Consider the relation schema Employee database.<br>EMPLOYEE (Fname ,Minit , Lname , <u>SSn</u> , Bdates , Address , Sex , Salary<br>Super_SSn , Dno)<br>DEPARTMENT (Dname , <u>Dnumber</u> , Mgr_SSn , Mgr_start_date)<br>PROJECT (Pname , <u>PNumber</u> , Plocation , Dnum)<br>WORKS_ON ( <u>Essn</u> , <u>Pno</u> , Hours)<br>DEPENDENT ( <u>Essn</u> , Dependent_name , sex, Bdate , Relationship)<br>Write relational algebra queries for the following :<br>i) Retrieve the name and address of all employees who work for the<br>‘Research’ department.<br>ii) List the names of all employees with 2 or more dependents.<br>iii) Find the names of employees who work on all the projects controlled<br>by department number 5.<br>iv) List the names of employees who have no dependents. | 8              | L3             | CO3 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| Module – 3     |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| Q.5            | a.             | What is the need for normalization? Explain second and third normal form with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6              | L2             | CO4 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
|                | b.             | Outline constraints in SQL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6              | L2             | CO1 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
|                | c.             | Identify the given Relation R(ABCDE) and its instance, check whether FDS given hold or not. Give reasons.<br>i) $A \rightarrow B$ ii) $B \rightarrow C$ iii) $D \rightarrow E$ iv) $CD \rightarrow E$ . <table border="1"><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>a<sub>1</sub></td><td>b<sub>1</sub></td><td>c<sub>1</sub></td><td>d<sub>1</sub></td><td>e<sub>1</sub></td></tr><tr><td>a<sub>1</sub></td><td>b<sub>2</sub></td><td>c<sub>1</sub></td><td>d<sub>1</sub></td><td>e<sub>1</sub></td></tr><tr><td>a<sub>2</sub></td><td>b<sub>2</sub></td><td>c<sub>1</sub></td><td>d<sub>2</sub></td><td>e<sub>3</sub></td></tr><tr><td>a<sub>2</sub></td><td>b<sub>3</sub></td><td>c<sub>3</sub></td><td>d<sub>2</sub></td><td>e<sub>2</sub></td></tr></table>          | A              | B              | C   | D | E | a <sub>1</sub> | b <sub>1</sub> | c <sub>1</sub> | d <sub>1</sub> | e <sub>1</sub> | a <sub>1</sub> | b <sub>2</sub> | c <sub>1</sub> | d <sub>1</sub> | e <sub>1</sub> | a <sub>2</sub> | b <sub>2</sub> | c <sub>1</sub> | d <sub>2</sub> | e <sub>3</sub> | a <sub>2</sub> | b <sub>3</sub> | c <sub>3</sub> | d <sub>2</sub> | e <sub>2</sub> | 8 | L3 | CO4 |
| A              | B              | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | D              | E              |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| a <sub>1</sub> | b <sub>1</sub> | c <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | d <sub>1</sub> | e <sub>1</sub> |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| a <sub>1</sub> | b <sub>2</sub> | c <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | d <sub>1</sub> | e <sub>1</sub> |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| a <sub>2</sub> | b <sub>2</sub> | c <sub>1</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | d <sub>2</sub> | e <sub>3</sub> |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| a <sub>2</sub> | b <sub>3</sub> | c <sub>3</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | d <sub>2</sub> | e <sub>2</sub> |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| OR             |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                |     |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
| Q.6            | a.             | What is Multivalued dependency? Explain 4NF and 5NF with suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6              | L2             | CO4 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
|                | b.             | Outline the informal design guidelines for relational schema.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6              | L2             | CO4 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |
|                | c.             | Consider relation R with following function dependency :<br>EMPPROJ ( <u>SSn</u> , <u>Pnumber</u> , Hours , Ename , Pname , Plocation)<br>SSN , Pnumber $\rightarrow$ Hours,<br>SSN $\rightarrow$ Ename<br>Pnumber $\rightarrow$ Pname , Plocation.<br>Is it 2NF? Verify? If no give reason.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8              | L3             | CO4 |   |   |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |                |   |    |     |

| Module – 4 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |           |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------|
| Q.7        | a. | Consider the following schema for a company database :<br>Employee (FName , LName , SSn , Address , Sex , Salary , Dno , Super_SSn)<br>Department (Dname , Dnumber , mgr_SSn , mgr_st_date)<br>Project (Pname , Pnumber , Plocation , Dnum)<br>WORKS_on (Essn , Pno , Hours)<br>DEPENDENT (Essn , Dependent name , Sex , Bdate , relationship).<br>Write the SQL queries for the following :<br>i) List the names of managers who have atleast one dependent (use correlated nested).<br>ii) Retrieve the name of each employee who has a dependent with the same first name and is the same sex as the employee.<br>iii) For each project retrieve the project number , project name and the number of employees who work on that project.<br>iv) Retrieve the SSN of all employees who work on project number 1, 2 or 3. (Use 1N).<br>v) Find the sum of the salaries of all employees of the 'Research' department as well as maximum salary , minimum salary , average salary in this department. | 10 | L3<br>CO3 |
|            | b. | Why concurrency control is needed? Demonstrate with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10 | L2<br>CO5 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |           |
| Q.8        | a. | Consider the following schedule. The actions are listed in the order they are scheduled and prefixed with the transaction name.<br>S1 : T1 : R(X) , T2 : R(X) T1 : W(Y) , T2 : W(Y) , T1 : R(Y) , T2 : R(Y)<br>S2 : T3 : W(X) , T1 : R(X) , T1 : W(Y) , T2 : R(Z) , T2 : W(Z) , T3 : R(Z)<br>For each schedule answer the following :<br>i) What is the precedence graph for the schedule?<br>ii) Is the schedule conflict_serializable? If so what are all the conflicts equivalent serial schedules?<br>iii) Is the schedule view serializable? If so what are all the view equivalent serial schedules?                                                                                                                                                                                                                                                                                                                                                                                            | 10 | L3<br>CO5 |
|            | b. | Explain triggers with example write a trigger in SQL to call a procedure "Inform_Supervisor" whenever an employees salary is greater than the salary of his or her direct supervisor in the COMPANY database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10 | L3<br>CO5 |
| Module – 5 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |           |
| Q.9        | a. | Describe the two – phase locking protocol for concurrency control provide example to illustrate how it ensures serializability in transaction schedule.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10 | L2<br>CO5 |
|            | b. | Explain the characteristics of NOSQL system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 | L2<br>CO6 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |           |
| Q.10       | a. | Explain binary locks and shared lock with algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 | L2<br>CO5 |
|            | b. | Explain MongoDB data model, CRUD operations and distributed system characteristics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10 | L2<br>CO6 |

## Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Database Management System

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | M         | L           | C        |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|----------|--|-------------|--|-----|------|-------|--------|--------|----------|---|-------|----|------|----|----|---|-----|----|------|----|----|---|-----|----|------|----|-------|-----|--------------|--------|-----|---------------|----|-----|--------------|----|-----|---------------|----|----|----|
| Q.1        | a.            | Define the following terms:<br>(i) Database (ii) Schema (iii) Entity<br>(iv) DDL (v) Degree of a relationship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 05        | L1          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | b.            | Briefly explain characteristics of database approach.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 05        | L2          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | c.            | List and explain advantages of using DBMS approach.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10        | L2          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| OR         |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| Q.2        | a.            | Define the following terms:<br>(i) Cardinality (ii) Weak entity (iii) Program data independence<br>(iv) DML (v) Value sets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05        | L1          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | b.            | Describe three-schema architecture. Why do we need mappings between schema levels?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 05        | L2          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | c.            | Explain different types of attributes in ER model with suitable example for each.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10        | L2          | CO1      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| Module – 2 |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| Q.3        | a.            | With suitable example, explain the entity integrity and referential integrity constraints. Why each is considered important?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 05        | L2          | CO2      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | b.            | Discuss equijoin and natural join with suitable example using relational algebra notation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05        | L2          | CO2      |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
|            | c.            | Given the relational tables:<br><table><tr><td colspan="4">Employee:</td><td colspan="2">Department:</td></tr><tr><td>EID</td><td>Name</td><td>DepID</td><td>Salary</td><td>DeptID</td><td>DeptName</td></tr><tr><td>1</td><td>Alice</td><td>10</td><td>5000</td><td>10</td><td>HR</td></tr><tr><td>2</td><td>Bob</td><td>20</td><td>6000</td><td>20</td><td>IT</td></tr><tr><td>3</td><td>Eve</td><td>20</td><td>6500</td><td>30</td><td>Sales</td></tr></table><br>Project<br><table><tr><td>PID</td><td>Project Name</td><td>DeptID</td></tr><tr><td>101</td><td>Project Alpha</td><td>10</td></tr><tr><td>102</td><td>Project Beta</td><td>20</td></tr><tr><td>103</td><td>Project Gamma</td><td>30</td></tr></table><br>Write relational algebra expression for the following:<br>(i) Find the names and salaries of all employees in the ‘IT’ department.<br>(ii) Find the ID’s and names of employees who are in the ‘IT’ department and have a salary greater than 6000.<br>(iii) Find the ID’s and names of employees who are either in the ‘HR’ department or have a salary greater than 6000.<br>(iv) Find the names of employees who are not in the ‘IT’ department<br>(v) Find the names of employees along with their department names. | Employee: |             |          |  | Department: |  | EID | Name | DepID | Salary | DeptID | DeptName | 1 | Alice | 10 | 5000 | 10 | HR | 2 | Bob | 20 | 6000 | 20 | IT | 3 | Eve | 20 | 6500 | 30 | Sales | PID | Project Name | DeptID | 101 | Project Alpha | 10 | 102 | Project Beta | 20 | 103 | Project Gamma | 30 | 10 | L3 |
| Employee:  |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           | Department: |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| EID        | Name          | DepID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Salary    | DeptID      | DeptName |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 1          | Alice         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5000      | 10          | HR       |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 2          | Bob           | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6000      | 20          | IT       |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 3          | Eve           | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6500      | 30          | Sales    |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| PID        | Project Name  | DeptID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 101        | Project Alpha | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 102        | Project Beta  | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |
| 103        | Project Gamma | 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |             |          |  |             |  |     |      |       |        |        |          |   |       |    |      |    |    |   |     |    |      |    |    |   |     |    |      |    |       |     |              |        |     |               |    |     |              |    |     |               |    |    |    |

1 of 3

## OR

| Q.4         | a.            | Explain any two operations that change the state of relation in a database. Provide suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 05       | L2 | CO2 |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|-----|------|---|-------|---|-----|---|-------|----------|--|-----|--------------|---|-------|---|------|---|-------|-----------|--|-----|---------------|---|--------|---|------|---|-----|-------------|--|-----|-----|---|---|---|---|---|---|---|---|----|----|-----|
|             | b.            | Discuss the aggregation functions and grouping in relational algebra with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 05       | L2 | CO2 |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
|             | c.            | <div>Given the relational tables:</div> <div><table><tr><th colspan="2">Student:</th></tr><tr><th>SID</th><th>Name</th></tr><tr><td>a</td><td>Alice</td></tr><tr><td>b</td><td>Bob</td></tr><tr><td>c</td><td>Carol</td></tr></table><table><tr><th colspan="2">Project:</th></tr><tr><th>PID</th><th>Project Name</th></tr><tr><td>p</td><td>Alpha</td></tr><tr><td>q</td><td>Beta</td></tr><tr><td>r</td><td>Gamma</td></tr></table><br/><table><tr><th colspan="2">Language:</th></tr><tr><th>LID</th><th>Language Name</th></tr><tr><td>x</td><td>Python</td></tr><tr><td>y</td><td>Java</td></tr><tr><td>z</td><td>C++</td></tr></table><table><tr><th colspan="2">Enrollment:</th></tr><tr><th>SID</th><th>PID</th></tr><tr><td>a</td><td>p</td></tr><tr><td>a</td><td>q</td></tr><tr><td>b</td><td>q</td></tr><tr><td>c</td><td>r</td></tr></table></div> <div>Write relational algebra expression for the following:</div> <div><div>(i) Rename the student table to Learner and display it.</div><div>(ii) Find the students (learners) who are not enrolled in any project.</div><div>(iii) Find the students who are enrolled in all projects.</div><div>(iv) Find the students who are not enrolled in any project.</div><div>(v) Find the students who are enrolled in both the ‘Alpha’ and ‘Beta’ projects.</div></div> | Student: |    | SID | Name | a | Alice | b | Bob | c | Carol | Project: |  | PID | Project Name | p | Alpha | q | Beta | r | Gamma | Language: |  | LID | Language Name | x | Python | y | Java | z | C++ | Enrollment: |  | SID | PID | a | p | a | q | b | q | c | r | 10 | L3 | CO2 |
| Student:    |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| SID         | Name          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| a           | Alice         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| b           | Bob           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| c           | Carol         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| Project:    |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| PID         | Project Name  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| p           | Alpha         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| q           | Beta          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| r           | Gamma         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| Language:   |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| LID         | Language Name |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| x           | Python        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| y           | Java          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| z           | C++           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| Enrollment: |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| SID         | PID           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| a           | p             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| a           | q             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| b           | q             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |
| c           | r             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |    |     |      |   |       |   |     |   |       |          |  |     |              |   |       |   |      |   |       |           |  |     |               |   |        |   |      |   |     |             |  |     |     |   |   |   |   |   |   |   |   |    |    |     |

## Module – 3

|     |    |                                                                                                                                                                                                                                                  |    |    |     |
|-----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.5 | a. | Explain Armstrong inference rules.                                                                                                                                                                                                               | 05 | L2 | CO4 |
|     | b. | What is the need for normalization? Explain 1NF, 2NF and 3NF with examples.                                                                                                                                                                      | 05 | L2 | CO4 |
|     | c. | What is functional dependency? Write an algorithm to find minimal cover for set of functional dependencies. Construct minimal cover M for set of functional dependencies which are: $E = \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$ | 10 | L3 | CO4 |

## OR

|     |    |                                                                                                                                                                                                                                |    |    |     |
|-----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6 | a. | Explain the types of update anomalies in SQL with an example.                                                                                                                                                                  | 05 | L2 | CO4 |
|     | b. | Explain types of JDBC drivers.                                                                                                                                                                                                 | 05 | L2 | CO5 |
|     | c. | Consider the schema $R = ABCD$ , subjected to FDs $F = \{A \rightarrow B, B \rightarrow C\}$ , and the non-binary partition $D1 = \{ACD, AB, BC\}$ . State whether D1 is a lossless decomposition? [give all steps in detail]. | 10 | L3 | CO4 |

## Module – 4

|     |    |                                                                    |    |    |     |
|-----|----|--------------------------------------------------------------------|----|----|-----|
| Q.7 | a. | Define transaction. Discuss ACID properties.                       | 05 | L2 | CO5 |
|     | b. | With a neat diagram, explain transition diagram of a transaction.  | 05 | L2 | CO5 |
|     | c. | Demonstrate working of assertion and triggers in SQL with example. | 10 | L3 | CO5 |

## OR

|     |    |                                                                                                                                                                                                                                                                                   |    |    |     |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.8 | a. | Explain cursor and its properties in embedded SQL with suitable example.                                                                                                                                                                                                          | 05 | L2 | CO5 |
|     | b. | <p>Determine if the following schedule is serializable and explain your reasoning:</p> <p>i) <math>T1 : R(X)W(X) \quad T2 : R(X)W(X) \quad T1 : COMMIT \quad T2 : COMMIT</math></p> <p>ii) <math>T1 : W(X)R(Y) \quad T2 : R(X)W(Y) \quad T1 : COMMIT \quad T2 : COMMIT</math></p> | 05 | L2 | CO5 |

|                   |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |            |
|-------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
|                   | <b>c.</b> | Consider the tables below:<br>Sailors ( <u>sid</u> : integer, sname : string, rating : integer, age : real)<br>Boats ( <u>bid</u> : integer, bname : string, color : string);<br>Reserves ( <u>sid</u> : integer, <u>bid</u> : integer, day : date)<br>Write SQL queries for the following:<br>(i) Write create table statement for reserves.<br>(ii) Find all information of sailors who have reserved boat number 101.<br>(iii) Find the names of sailors who have reserved at least one boat.<br>(iv) Find the names of sailors who have reserved a red boat.<br>(v) Find the average age of sailors for each rating level. | <b>10</b> | <b>L3</b> | <b>CO5</b> |
| <b>Module – 5</b> |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |            |
| <b>Q.9</b>        | <b>a.</b> | Explain the CAP theorem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>05</b> | <b>L2</b> | <b>CO6</b> |
|                   | <b>b.</b> | What is NOSQL graph database? Explain Neo4j.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>05</b> | <b>L2</b> | <b>CO6</b> |
|                   | <b>c.</b> | Why concurrency control and recovery are needed in DBMS? Demonstrate with suitable examples types of problems that may occur when two simple transactions run concurrently.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>10</b> | <b>L3</b> | <b>CO5</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |           |            |
| <b>Q.10</b>       | <b>a.</b> | Explain basic operations CRUD in MongoDB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>05</b> | <b>L2</b> | <b>CO6</b> |
|                   | <b>b.</b> | Explain deadlock prevention protocols.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>05</b> | <b>L2</b> | <b>CO5</b> |
|                   | <b>c.</b> | Briefly discuss the two-phase locking techniques for concurrency control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>10</b> | <b>L3</b> | <b>CO5</b> |

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## Fourth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Database Management Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks, L: Bloom's level, C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          | M  | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Define Database. Explain the characteristics of database.                                                                                                                                                                                                                                                                                                                                                                                                | 06 | L2 | CO1 |
|            | b. | Discuss three schema architecture with a diagram and explain the importance of mappings between the schema levels.                                                                                                                                                                                                                                                                                                                                       | 08 | L2 | CO1 |
|            | c. | Discuss the types of end users with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                   | 06 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |     |
| Q.2        | a. | Illustrate the component modulus of DBMS and their interaction with a neat diagram.                                                                                                                                                                                                                                                                                                                                                                      | 06 | L2 | CO1 |
|            | b. | List and explain different types of attributes with example.                                                                                                                                                                                                                                                                                                                                                                                             | 04 | L2 | CO1 |
|            | c. | Draw an ER-diagram for keeping track of information about 'Employee' database, taking into accounts at least five entities.                                                                                                                                                                                                                                                                                                                              | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |     |
| Q.3        | a. | Define the following terms:<br>i) Primary key    ii) Super key    iii) Candidate key    iv) Foreign key                                                                                                                                                                                                                                                                                                                                                  | 04 | L1 | CO2 |
|            | b. | Discuss the various constraints violations during insert, delete and update operations with examples for each.                                                                                                                                                                                                                                                                                                                                           | 08 | L3 | CO2 |
|            | c. | Explain binary relational algebra operations with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                     | 08 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |     |
| Q.4        | a. | Summarize the E-R to Relational Mapping algorithm with examples for each step.                                                                                                                                                                                                                                                                                                                                                                           | 07 | L3 | CO2 |
|            | b. | Discuss the various types of JOIN operation with an examples.                                                                                                                                                                                                                                                                                                                                                                                            | 05 | L2 | CO2 |
|            | c. | Consider the following system:<br>SAILORS(Sid, Sname, rating, age)<br>BOATS(Bid, Bname, Bcolor)<br>RESERVES(Sid, Bid, day)<br><br>Obtain the relational algebra queries for the following :<br>i) Find the name of sailors who reserved green boat.<br>ii) Find the color of the boat reserve by "Naresh"<br>iii) Find the name of the sailor who has reserved boat no. 1.<br>iv) Find the Sid of sailors with age over 20 who have not reserved a boat. | 08 | L3 | CO2 |
| 1 of 2     |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |     |

**Module – 3**

|            |           |                                                                  |           |           |            |
|------------|-----------|------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.5</b> | <b>a.</b> | Explain informal design guidelines for relational schema design. | <b>04</b> | <b>L2</b> | <b>CO3</b> |
|            | <b>b.</b> | Define Normalization. Explain with examples 1NF, 2NF, 3NF.       | <b>08</b> | <b>L3</b> | <b>CO3</b> |
|            | <b>c.</b> | Discuss the types of update anomalies in SQL with an example.    | <b>08</b> | <b>L3</b> | <b>CO3</b> |

**OR**

|            |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |           |            |
|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.6</b> | <b>a.</b> | Explain with suitable example how constraints are specified in SQL during table creation.                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>10</b> | <b>L3</b> | <b>CO3</b> |
|            | <b>b.</b> | Consider the following schema:<br>Employee (SSN, F_name, M_name, L_name, Address, Salary, D_no)<br>Department (Dname, Dno, MgrSSN, Mgr_S_Date)<br>i) Retrieve the name and address of all employees who work for the 'Research' Department.<br>ii) Retrieve all employees whose address is in Houston, Texas.<br>iii) Retrieve all employees in department 5 whose salary is between \$30,000 and \$40,000.<br>iv) Update all employees in the 'Research' department a raise in salary of 10%.<br>v) Delete employee having SSN = 123456789; | <b>10</b> | <b>L3</b> | <b>CO3</b> |

**Module – 4**

|            |           |                                                                                 |           |           |            |
|------------|-----------|---------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.7</b> | <b>a.</b> | Discuss the importance of concurrency control and give an example for the same. | <b>06</b> | <b>L2</b> | <b>CO4</b> |
|            | <b>b.</b> | Explain the states in transaction processing with a neat diagram.               | <b>08</b> | <b>L2</b> | <b>CO4</b> |
|            | <b>c.</b> | Discuss the ACID properties of database transaction.                            | <b>06</b> | <b>L2</b> | <b>CO4</b> |

**OR**

|            |           |                                                                                                                                                                                                                                   |           |           |            |
|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.8</b> | <b>a.</b> | What is the difference between 'where' and 'having' clause? Give an example.                                                                                                                                                      | <b>04</b> | <b>L2</b> | <b>CO4</b> |
|            | <b>b.</b> | Write a note on the following with an example :<br>i) Assertion                      ii) Triggers                                                                                                                                 | <b>06</b> | <b>L2</b> | <b>CO4</b> |
|            | <b>c.</b> | Create cursor for employee table and extract the values from the table. Declare the variable open the cursor and extract the values from the cursor. Close the cursor for the given relation Employee(E_id, E_Name, Age, Salary). | <b>10</b> | <b>L3</b> | <b>CO4</b> |

**Module – 5**

|            |           |                                                                      |           |           |            |
|------------|-----------|----------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.9</b> | <b>a.</b> | Discuss the two phase locking protocol used for concurrency control. | <b>08</b> | <b>L2</b> | <b>CO5</b> |
|            | <b>b.</b> | Explain Multiversion concurrency control techniques in detail.       | <b>08</b> | <b>L2</b> | <b>CO5</b> |
|            | <b>c.</b> | Write a note on intension locking used to achieve granularity.       | <b>04</b> | <b>L2</b> | <b>CO5</b> |

**OR**

|             |           |                                                                          |           |           |            |
|-------------|-----------|--------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.10</b> | <b>a.</b> | Explain the characteristic of NOSQL systems.                             | <b>08</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>b.</b> | What are the basic operations of CRUD in MongoDB?                        | <b>06</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>c.</b> | Write a note on Neo4j Interfaces and Distributed System Characteristics. | <b>06</b> | <b>L2</b> | <b>CO5</b> |

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